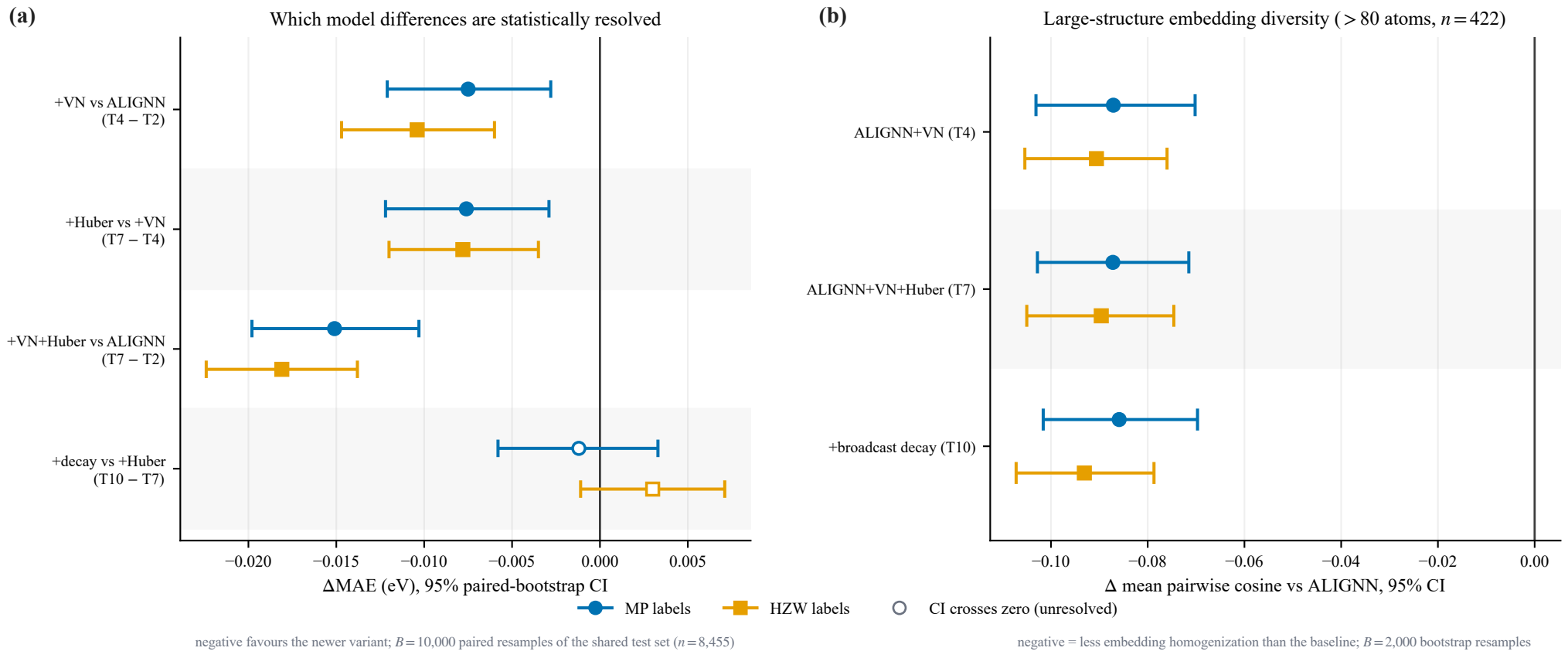

(a)
Which model differences are statistically resolved
+VN vs ALIGNN (T4 − T2)
+Huber vs +VN (T7 − T4)
+VN+Huber vs ALIGNN (T7 − T2)
+decay vs +Huber (T10 − T7)
−0.020
−0.015
−0.010
−0.005
0.000
0.005
ΔMAE (eV), 95% paired-bootstrap CI
negative favours the newer variant; B = 10,000 paired resamples of the shared test set (n = 8,455)
(b)
Large-structure embedding diversity (> 80 atoms, n = 422)
ALIGNN+VN (T4)
ALIGNN+VN+Huber (T7)
+broadcast decay (T10)
−0.10
−0.08
−0.06
−0.04
−0.02
0.00
Δ mean pairwise cosine vs ALIGNN, 95% CI
negative = less embedding homogenization than the baseline; B = 2,000 bootstrap resamples
MP labels
HZW labels
CI crosses zero (unresolved)